

TE Curve Verification Pipeline Familywise ONNX Report - wave4_3_mixture_density_k3

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k3/2026-07-15-17-24-28__track2_wave4_3_mixture_density_k3_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k3/2026-07-15-17-24-28__track2_wave4_3_mixture_density_k3_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k3/2026-07-15-17-24-28__track2_wave4_3_mixture_density_k3_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k3/2026-07-15-17-24-28__track2_wave4_3_mixture_density_k3_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave4_3_mixture_density_k3` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave4_3_mixture_density_k3_fw_simplified_setpoints</code>	<code>2026-07-15-10-35-14__te_wave4_3_mixture_density_k3_fw__simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave4_3_mixture_density_k3_bw_simplified_setpoints</code>	<code>2026-07-15-10-57-54__te_wave4_3_mixture_density_k3_bw__simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave4_3_mixture_density_k3_global_simplified_setpoints</code>	<code>2026-07-15-10-10-10__te_wave4_3_mixture_density_k3_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k3/forward/2026-07-15-10-35-14__te_wave4_3_mixture_density_k3_fw__simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k3/forward/2026-07-15-10-35-14__te_wave4_3_mixture_density_k3_fw__simplified_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=077-val_mae=0.00357399.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k3/backward/2026-07-15-10-57-54__te_wave4_3_mixture_density_k3_bw__simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k3/backward/2026-07-15-10-57-54__te_wave4_3_mixture_density_k3_bw__simplified_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=083-val_mae=0.00361315.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k3/global/2026-07-15-10-10-10__te_wave4_3_mixture_density_k3_global__simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k3/global/2026-07-15-10-10-10__te_wave4_3_mixture_density_k3_global__simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=103-val_mae=0.00358189.ckpt</code>

Aggregate Metrics

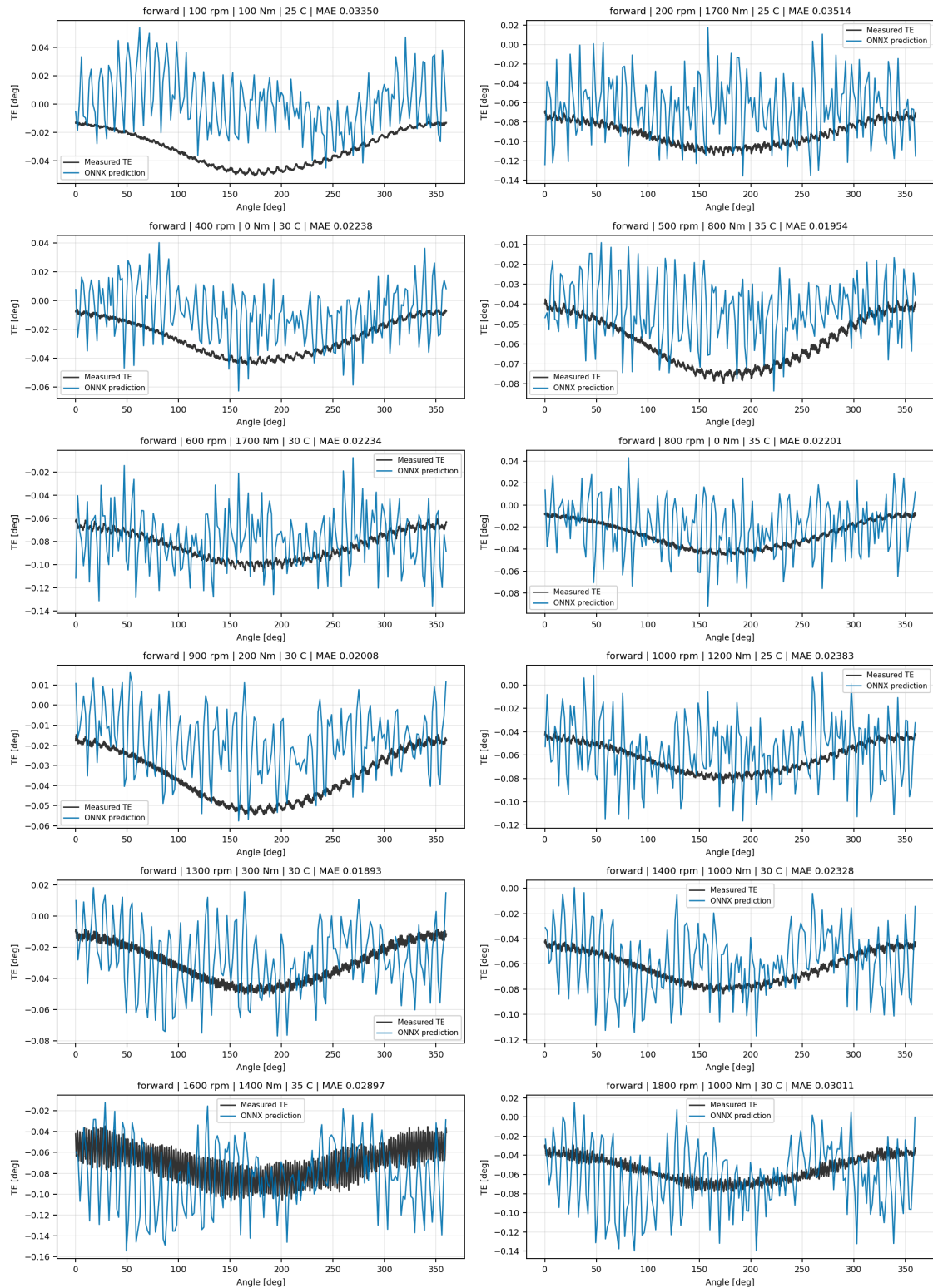
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.024845	0.030222	58.997	82.535
backward	97	0.034812	0.043383	82.585	108.227
global	194	0.021445	0.026134	50.907	76.902

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.008848	0.011949	0.021185	0.076285
backward	-0.019591	0.019631	0.030673	0.130940
global	-0.000115	0.014588	0.016652	0.080630

Forward 12-Curve Page

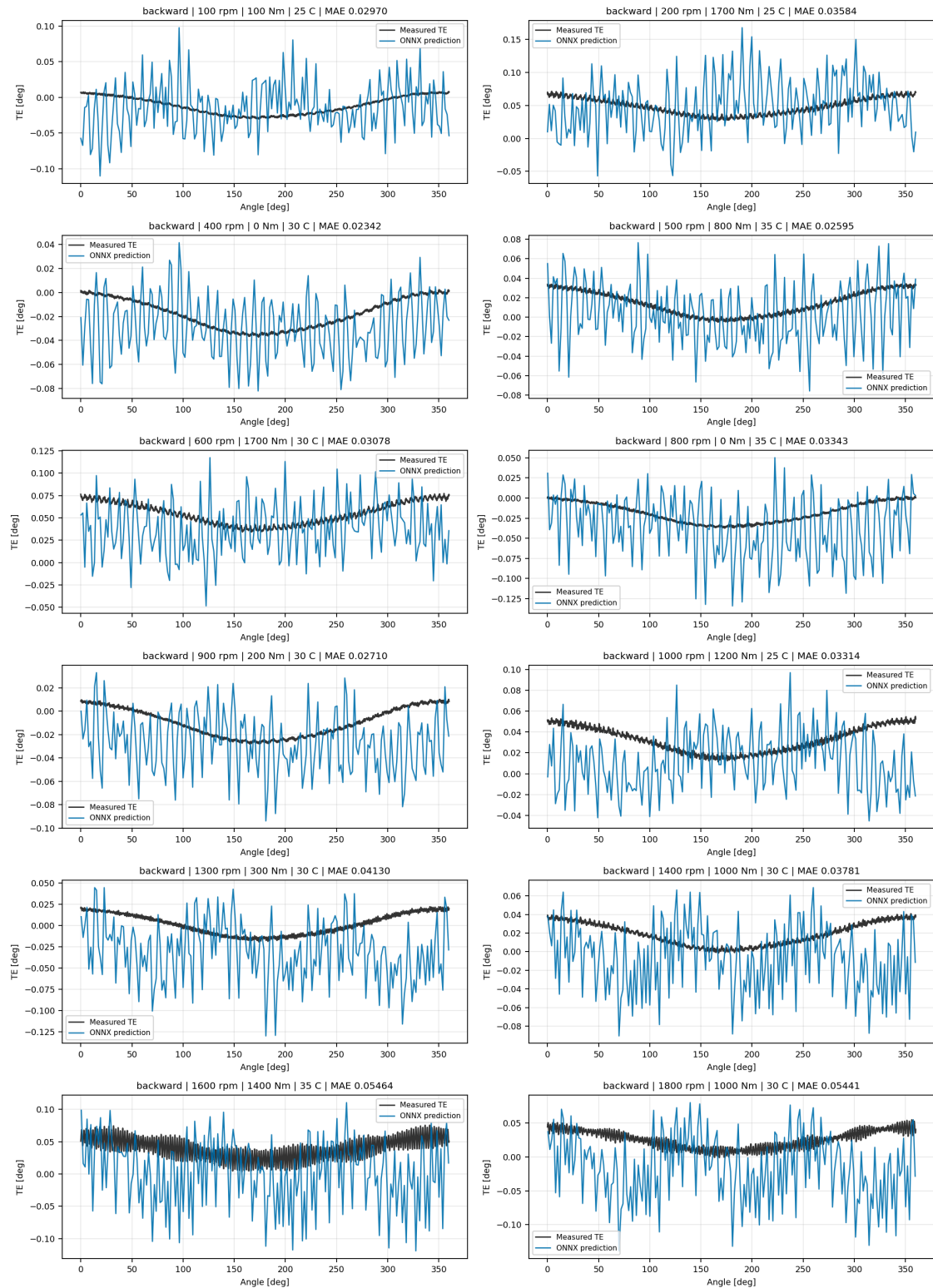
Simplified Dataset + Setpoints | wave4_3_mixture_density_k3 | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

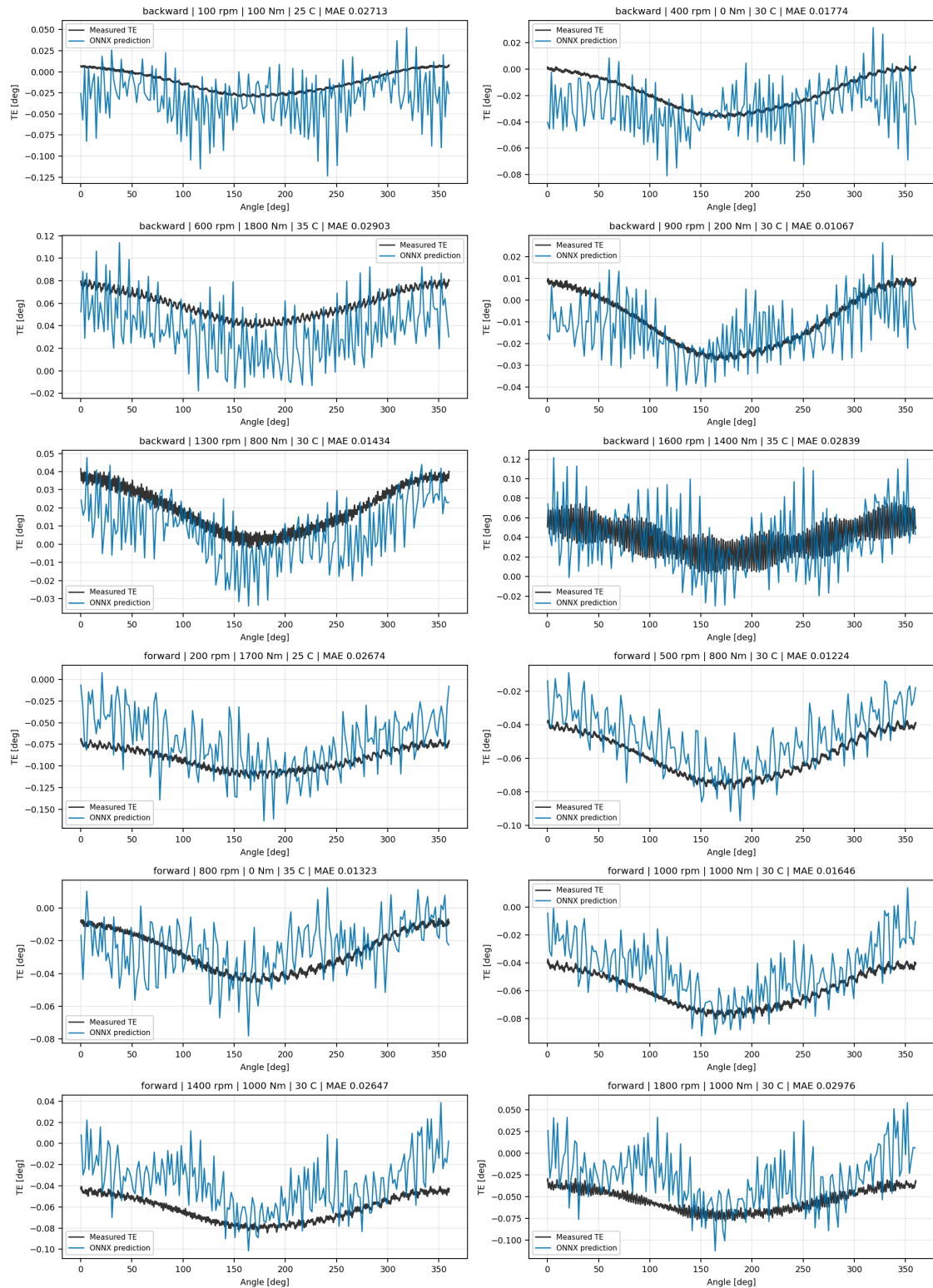
Simplified Dataset + Setpoints | wave4_3_mixture_density_k3 | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave4_3_mixture_density_k3 | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave4_3_mixture_density_k3 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_3_mixture_density_k3_fw_polished_setpoints	2026-07-15-12-21-49_te_wave4_3_mixture_density_k3_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave4_3_mixture_density_k3_bw_polished_setpoints	2026-07-15-12-55-40_te_wave4_3_mixture_density_k3_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave4_3_mixture_density_k3_global_polished_setpoints	2026-07-15-11-41-10_te_wave4_3_mixture_density_k3_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k3/forward/2026-07-15-12-21-49_te_wave4_3_mixture_density_k3_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k3/forward/2026-07-15-12-21-49_te_wave4_3_mixture_density_k3_fw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=140-val_mae=0.00184649.ckpt
backward	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k3/backward/2026-07-15-12-55-40_te_wave4_3_mixture_density_k3_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k3/backward/2026-07-15-12-55-40_te_wave4_3_mixture_density_k3_bw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=166-val_mae=0.00183407.ckpt
global	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k3/global/2026-07-15-11-41-10_te_wave4_3_mixture_density_k3_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k3/global/2026-07-15-11-41-10_te_wave4_3_mixture_density_k3_global_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=177-val_mae=0.00183826.ckpt

Aggregate Metrics

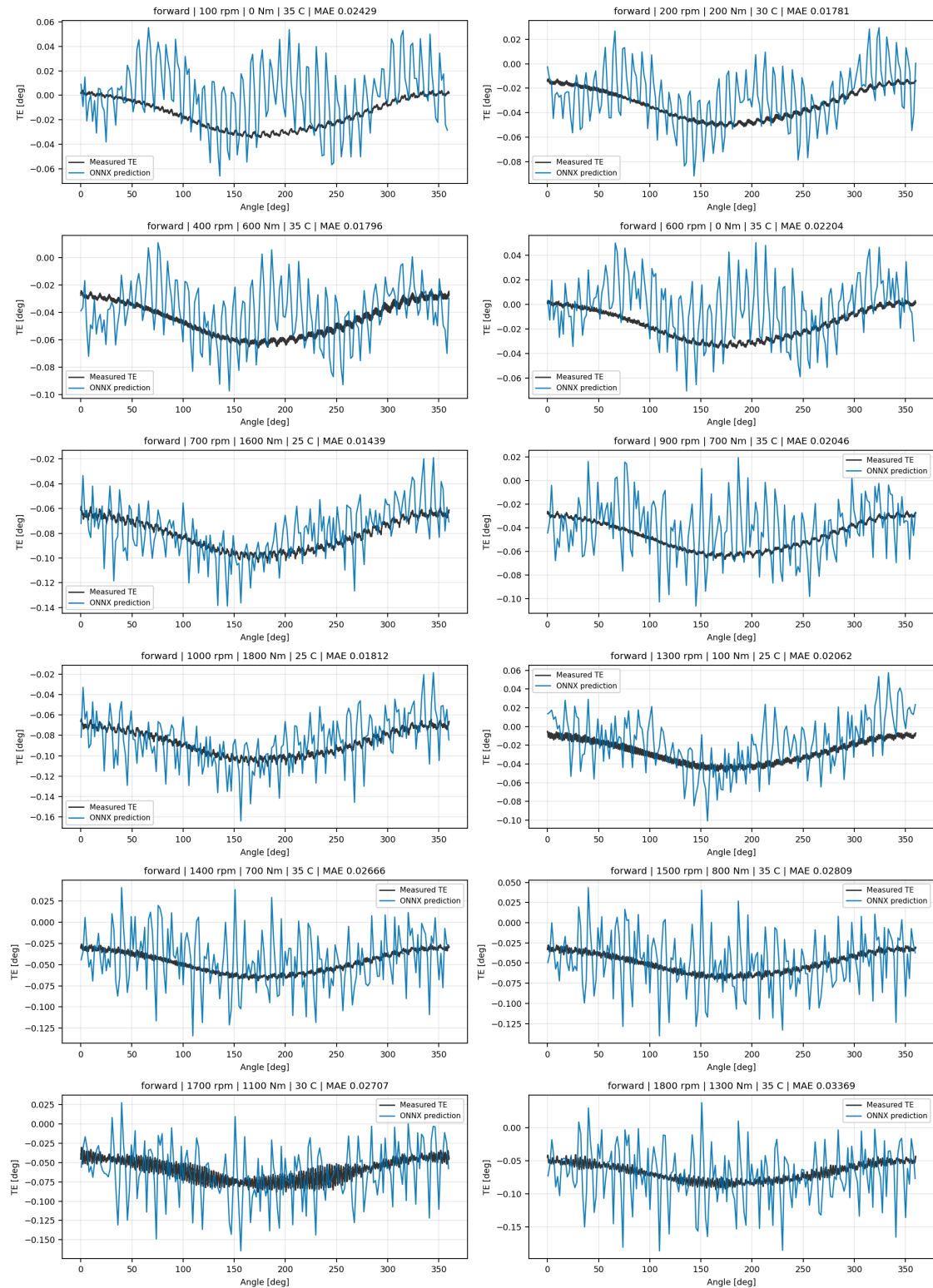
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.020326	0.025385	47.446	68.898
backward	94	0.018191	0.022434	41.698	61.031
global	194	0.025600	0.031621	59.389	86.511

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.004207	0.004816	0.019724	0.099689
backward	-0.009033	0.009637	0.016173	0.074918
global	0.000349	0.012429	0.022387	0.095416

Forward 12-Curve Page

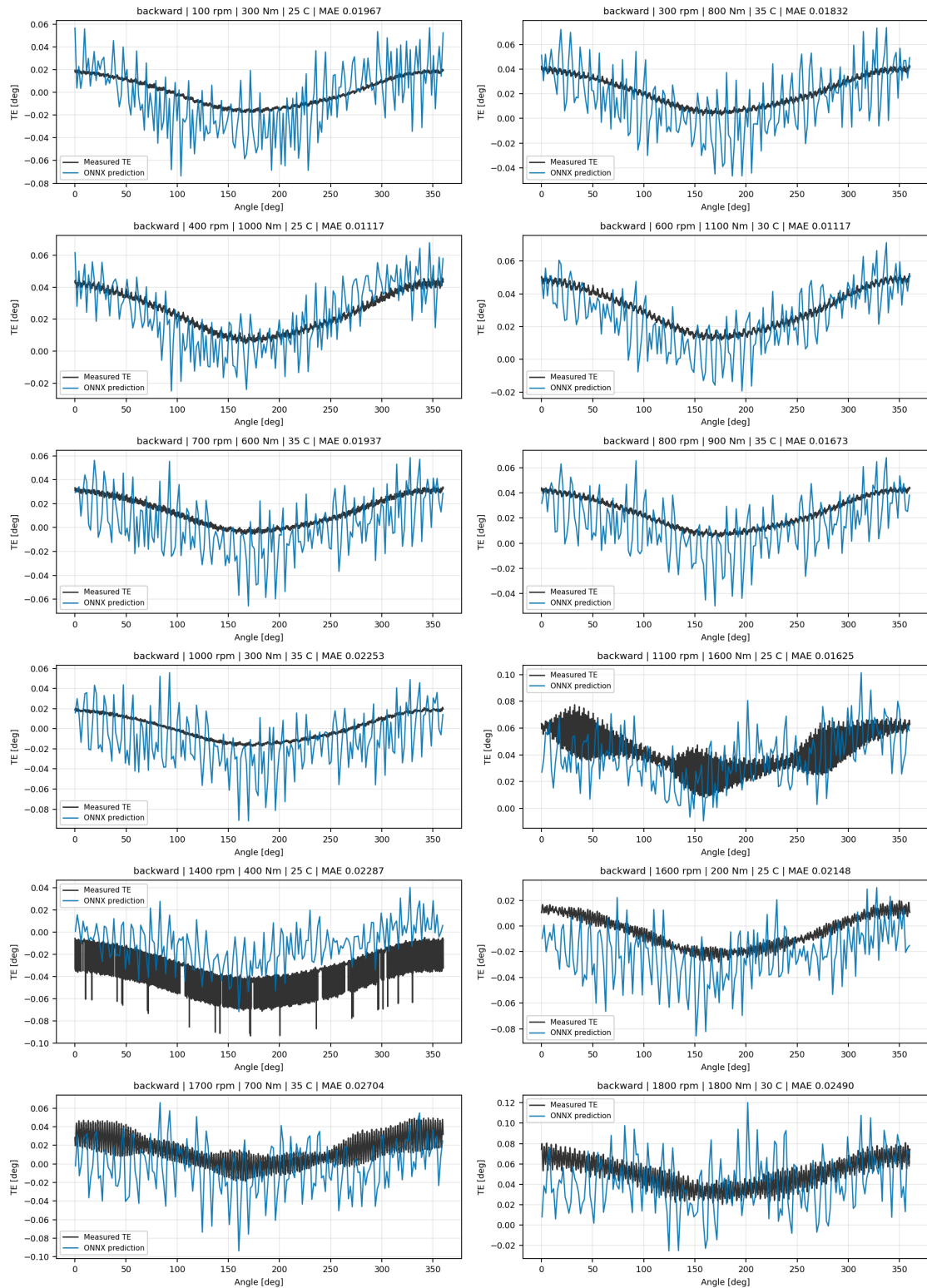
Polished Dataset + Setpoints | wave4_3_mixture_density_k3 | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

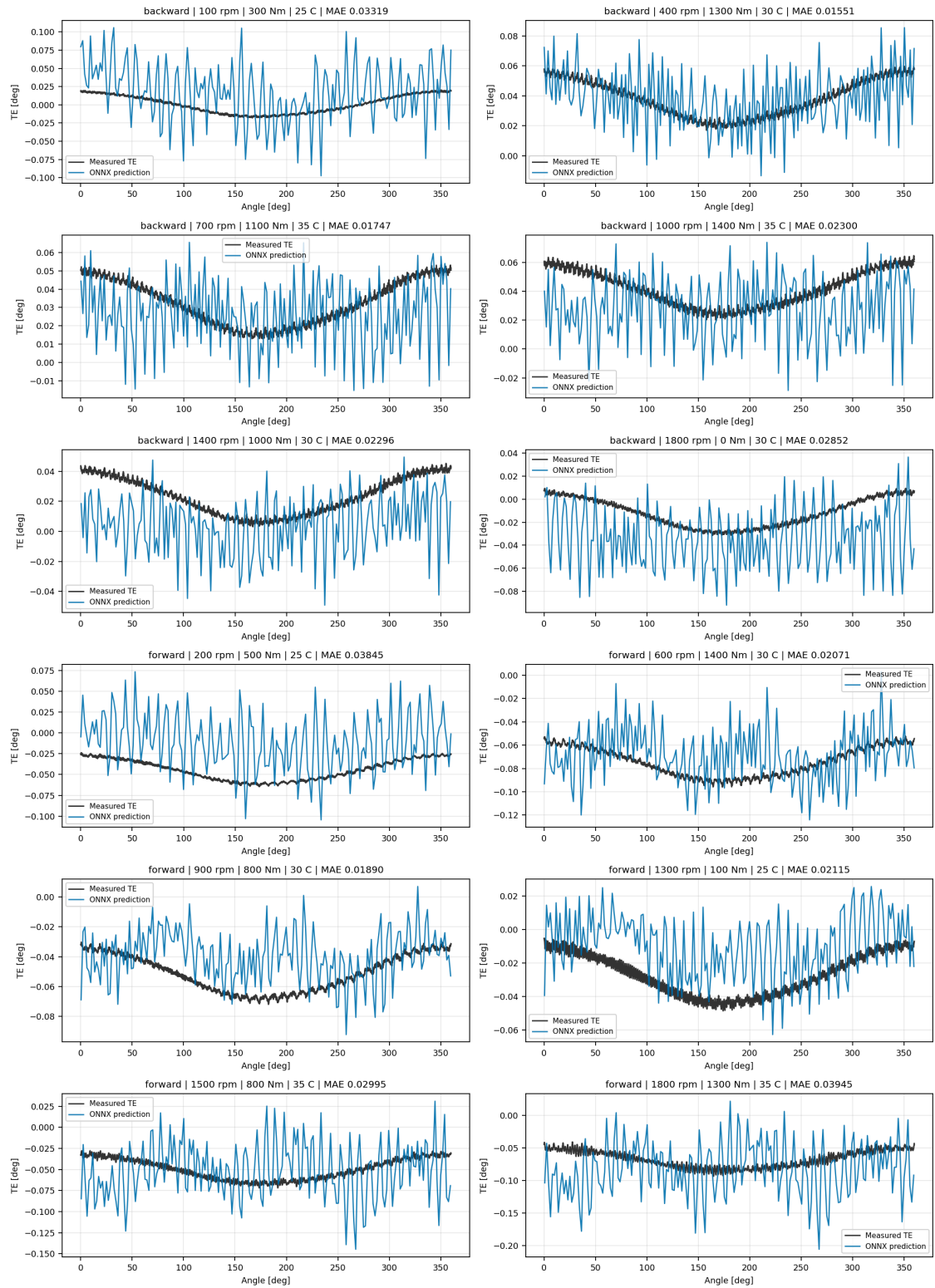
Polished Dataset + Setpoints | wave4_3_mixture_density_k3 | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave4_3_mixture_density_k3 | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave4_3_mixture_density_k3 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_3_mixture_density_k3_fw_polished_actual_values	2026-07-15-14-39-00__te_wave4_3_mixture_density_k3_fw__polished_actual_values	polished_point_v1
backward	te_wave4_3_mixture_density_k3_bw_polished_actual_values	2026-07-15-15-39-35__te_wave4_3_mixture_density_k3_bw__polished_actual_values	polished_point_v1
global	te_wave4_3_mixture_density_k3_global_polished_actual_values	2026-07-15-13-58-20__te_wave4_3_mixture_density_k3_global__polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k3/forward/2026-07-15-14-39-00__te_wave4_3_mixture_density_k3_fw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k3/forward/2026-07-15-14-39-00__te_wave4_3_mixture_density_k3_fw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=240-val_mae=0.00161545.ckpt
backward	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k3/backward/2026-07-15-15-39-35__te_wave4_3_mixture_density_k3_bw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k3/backward/2026-07-15-15-39-35__te_wave4_3_mixture_density_k3_bw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=076-val_mae=0.00181385.ckpt
global	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k3/global/2026-07-15-13-58-20__te_wave4_3_mixture_density_k3_global__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k3/global/2026-07-15-13-58-20__te_wave4_3_mixture_density_k3_global__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=164-val_mae=0.00178681.ckpt

Aggregate Metrics

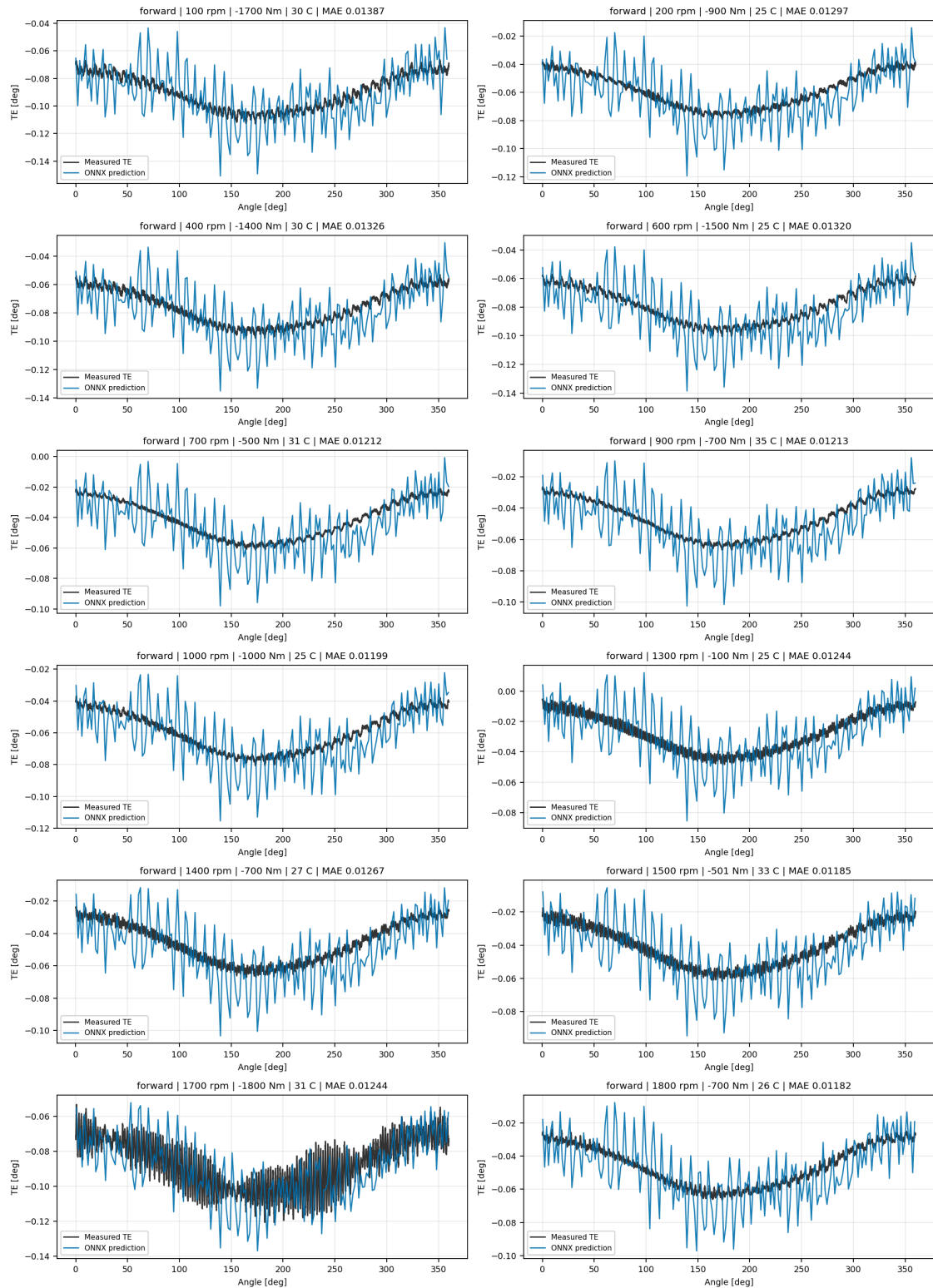
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.012587	0.015597	29.635	33.472
backward	94	0.031030	0.038352	71.999	85.121
global	194	0.011910	0.014642	27.766	39.988

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.004018	0.004018	0.012086	0.052961
backward	-0.014403	0.014742	0.028092	0.158233
global	0.000965	0.009109	0.009046	0.036483

Forward 12-Curve Page

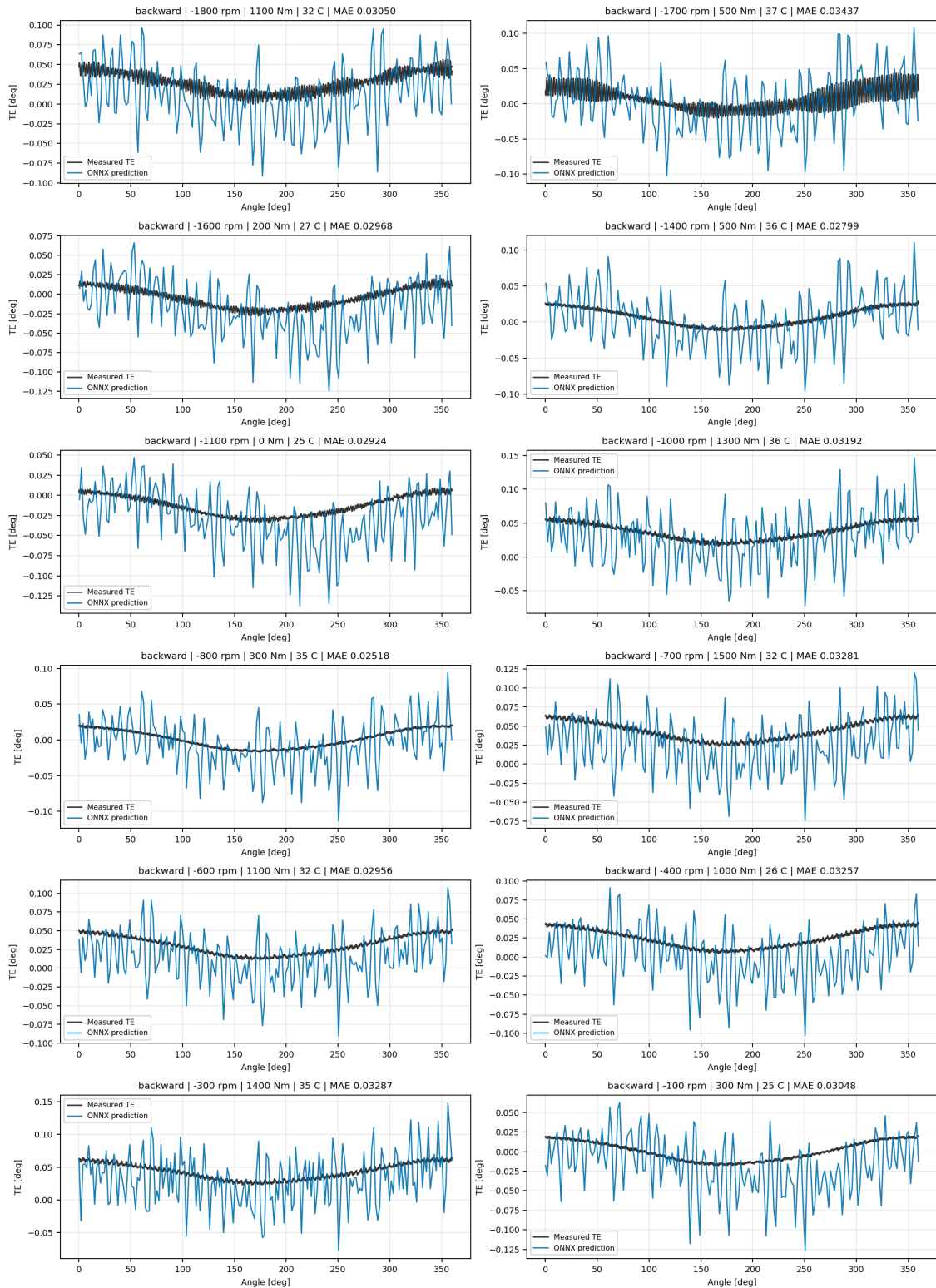
Polished Dataset + Actual Values | wave4_3_mixture_density_k3 | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

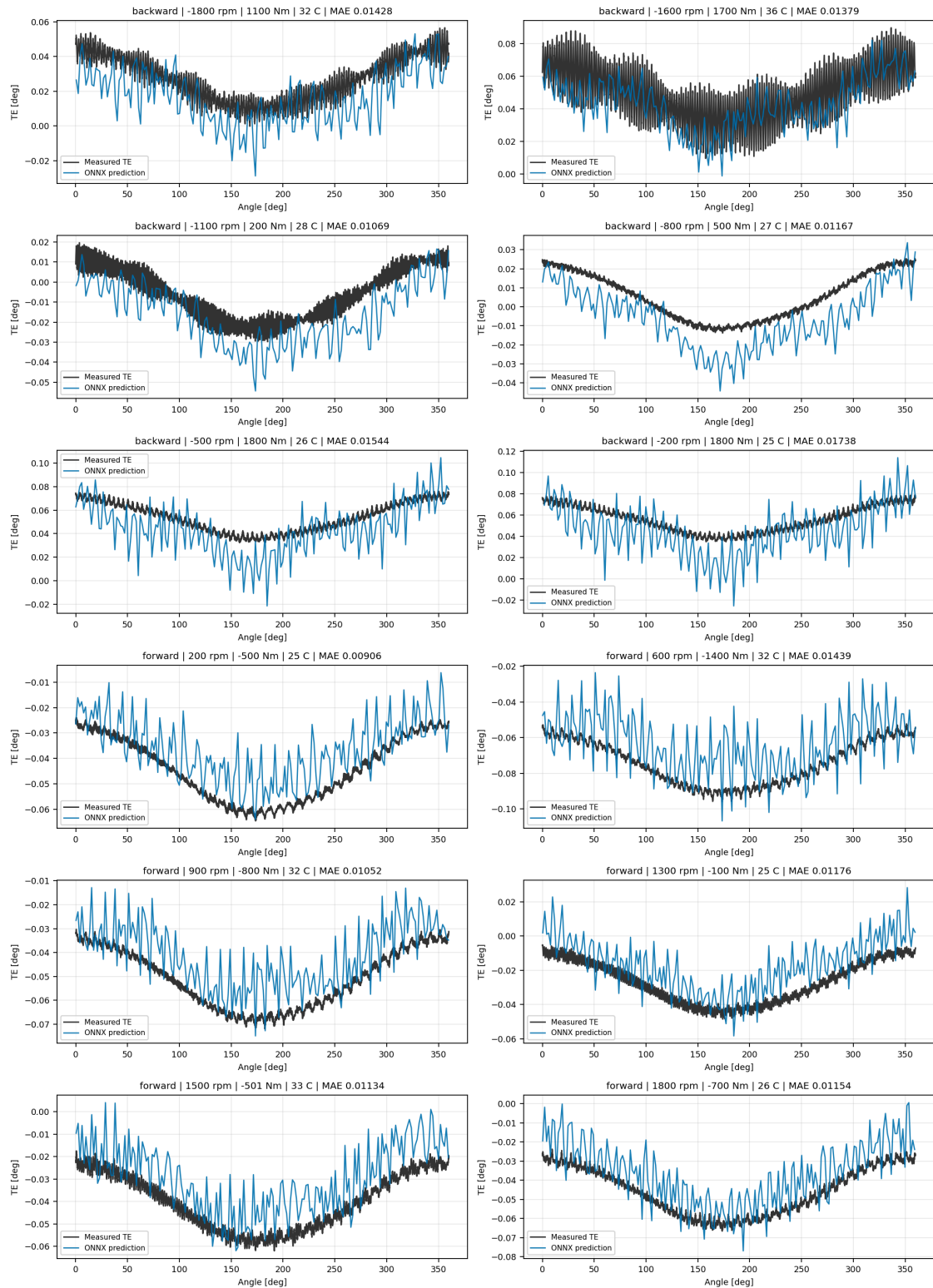
Polished Dataset + Actual Values | wave4_3_mixture_density_k3 | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave4_3_mixture_density_k3 | global



polished_dataset_actual_values global 12-curve collage